

3.4.2 Materials

3.4.2.1 Bulk properties of solids

Content	Opportunities for skills development
Density, $\rho = \frac{m}{V}$ Hooke's law, elastic limit, $F = k\Delta L$, k as stiffness and spring constant. Tensile strain and tensile stress. Elastic strain energy, breaking stress. $\text{energy stored} = \frac{1}{2}F\Delta L$ $= \text{area under force-extension graph}$ Description of plastic behaviour, fracture and brittle behaviour linked to force-extension graphs. Quantitative and qualitative application of energy conservation to examples involving elastic strain energy and energy to deform. Spring energy transformed to kinetic and gravitational potential energy. Interpretation of simple stress-strain curves. Appreciation of energy conservation issues in the context of ethical transport design.	MS 0.2, 4.3 / PS 3.3, 4.1 Students can compare the use of analogue and digital meters. MS 0.4, 4.3 / AT e Estimate the volume of an object leading to an estimate of its density.

3.4.2.2 The Young modulus

Content	Opportunities for skills development
$\text{Young modulus} = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{FL}{A\Delta L}$ Use of stress–strain graphs to find the Young modulus. (One simple method of measurement is required.)	MS 3.1
Required practical 4: Determination of the Young modulus by a simple method.	